

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.1: Trigonometry I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Trigonometry I
Driving Question	DRIVING QUESTION: How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Right-Angled Triangles and the Anatomy of an Angle: Setting Up the Shadow Problem	Teacher should display a photograph of a tall acacia tree in the Rift Valley casting a long shadow on a sunny afternoon. Students observe that the sunbeam, the tree trunk, and the shadow on the ground form a right-angled triangle. Teacher records all student noticings and wonderings on the Driving Question Board. Pedagogical note: Cold-call at least four students before accepting volunteer answers. Allow 30 seconds of silent thinking after posing 'What shape do you see hidden in this scene?'	Students learned that every right-angled triangle contains one 90° angle and two acute angles, and that the sides are named relative to a chosen acute angle θ : the side directly across from θ is the Opposite, the side next to θ (that is not the hypotenuse) is the Adjacent, and the longest side (across from 90°) is the Hypotenuse. They also reviewed that the three interior angles of any triangle sum to 180° , confirming that the two acute angles in a right-angled triangle always sum to 90° .	The shadow scenario is explained by the hidden right-angled triangle: the tree trunk is the Opposite side, the shadow is the Adjacent side, and the sunbeam travelling from the treetop to the shadow tip is the Hypotenuse. The angle the sunbeam makes with the ground is θ . Teacher should emphasise that correctly labelling Opposite, Adjacent, and Hypotenuse relative to θ — not a fixed position — is the foundational skill on which all three trigonometric ratios depend. Pedagogical note: A common misconception is that Hypotenuse changes with θ ; clarify that the Hypotenuse is always opposite the right angle and never changes, but Opposite and Adjacent swap when θ shifts to the other acute angle.
Lesson 2 Sine of an Acute Angle: Reading the Language of Triangles	Teacher guides students to draw three right-angled triangles of different sizes but all containing a 35° angle, measure the Opposite and Hypotenuse sides with a ruler, and compute Opposite $\div$ Hypotenuse for each. Students observe that despite the triangles being different sizes, the ratio is approximately the same (~ 0.574) every	The sine of an acute angle θ is defined as $\sin \theta = \text{Opposite} \div \text{Hypotenuse}$. This ratio is a fixed value that depends only on the angle, not on the size of the triangle. For the sunbeam problem, if the angle of elevation of the sun is known, $\sin \theta$ immediately links the tree height (Opposite) to the length of the sunbeam (Hypotenuse).	Because all right-angled triangles containing the same angle are similar (AA similarity), their corresponding sides are proportional, making Opposite $\div$ Hypotenuse invariant. This is why a table of sine values compiled once — like the mathematical tables in Kenyan secondary school log books — works for every triangle

	time. Pedagogical note: Insist students record measurements to the nearest millimetre and compute the ratio to three decimal places before the class pools results; this makes the invariance of the ratio genuinely surprising.		everywhere. Teacher should connect back to the driving question: if a surveyor in Kericho knows the angle of elevation of a hilltop and can measure the straight-line distance to it (the Hypotenuse), $\sin \theta \times \text{Hypotenuse}$ gives the hill's height directly.
Lesson 3 Cosine of an Acute Angle: The Shadow's Hidden Ratio	Using the same three triangles drawn in Lesson 2, students now measure the Adjacent side and compute $\text{Adjacent} \div \text{Hypotenuse}$ for each triangle with $\theta = 35^\circ$. They record results and observe that this new ratio is also constant across all three triangles (~ 0.819). Teacher then asks students to compare the sine value from Lesson 2 with this new ratio and pose the question: 'Are these the same number?' Pedagogical note: Allow 45 seconds of pair discussion before cold-calling three pairs.	The cosine of an acute angle θ is defined as $\cos \theta = \text{Adjacent} \div \text{Hypotenuse}$. Like sine, it is invariant for a given angle across all similar right-angled triangles. For the sunbeam scenario, $\cos \theta$ links the length of the shadow on the ground (Adjacent) to the length of the sunbeam (Hypotenuse): $\text{shadow length} = \cos \theta \times \text{Hypotenuse}$.	Cosine completes the 'Hypotenuse family' of ratios alongside sine. Teacher should highlight the memory aid SOH–CAH–TOA, writing it prominently and asking students to decode it aloud. A key pedagogical point: $\cos \theta$ for a given angle is always less than 1 (since $\text{Adjacent} < \text{Hypotenuse}$), and as θ increases from 0° toward 90° , the Adjacent shrinks, so $\cos \theta$ decreases — the opposite behaviour to sine. Teacher should pose: 'At midday in Nairobi when the sun is nearly overhead, is the shadow (Adjacent) long or short? What does that tell us about $\cos \theta$ when θ is large?' This anchors cosine behaviour to lived Kenyan experience.
Lesson 4 Tangent and the Sine–Cosine–Tangent Relationship: Solving the Mango-Tree Problem	Teacher presents a scenario: a student in Kisumu stands 8 m from a mango tree and measures the angle of elevation to the top as 42° . Students are asked to find the tree's height using only a shadow-length measurement (no sunbeam length available). Through guided discovery, students compute $\text{Opposite} \div \text{Adjacent}$ from their Lesson 2/3 triangles and observe a third constant ratio (~ 0.700 for 35°). Pedagogical note: Pose 'Which two sides do we know or can measure on the ground — without a ladder or drone?' before revealing the tangent definition.	Learners learned that $\tan \theta = \text{Opposite} \div \text{Adjacent}$ and that this ratio is algebraically equivalent to $\sin \theta \div \cos \theta$, making all three trigonometric ratios interconnected. Tangent is the most directly useful ratio for the driving question because it links height (Opposite) directly to shadow length (Adjacent) — both measurable on the ground without climbing.	The algebraic identity $\tan \theta = \sin \theta / \cos \theta$ emerges naturally: $(\text{Opp}/\text{Hyp}) \div (\text{Adj}/\text{Hyp}) = \text{Opp}/\text{Adj}$. Teacher should work through the mango-tree problem: $\text{height} = \tan 42^\circ \times 8 \text{ m} \approx 0.900 \times 8 = 7.2 \text{ m}$. Emphasise that $\tan \theta$ is the ratio that directly answers the driving question when the horizontal distance (shadow or baseline) is measurable. Pedagogical note: A common error is inverting the ratio ($\text{Adjacent} \div \text{Opposite}$); reinforce SOH–CAH–TOA and insist students label sides before writing any ratio.
Lesson 5 Trigonometric Ratios Using Calculators: Precision, Inverse Functions, and the Shadow Puzzle	Students use scientific calculators (or the ARES offline calculator tool) to evaluate $\sin 35^\circ$, $\cos 35^\circ$, and $\tan 35^\circ$, comparing results to their hand-drawn measurements from Lessons 2–3. They then use the inverse functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) to recover the angle from a known ratio. Teacher poses: 'If a Nairobi skyscraper casts a 50 m shadow and the sunbeam angle is unknown, but you measure the building is 120 m tall, how	Learners discovered that the calculator returns consistent, high-precision trigonometric ratio values that match but extend beyond hand-drawn approximations. They also learned that inverse trigonometric functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) reverse the process: given a ratio, they return the angle. This allows surveyors to find unknown angles from measured side lengths.	The calculator is a compressed table of all trigonometric ratios computed to many decimal places. Teacher should model the full two-way workflow: (1) angle $\rightarrow$ ratio (forward: press $\sin/\cos/\tan$), (2) ratio $\rightarrow$ angle (inverse: press $\text{SHIFT} + \sin/\cos/\tan$). For the skyscraper problem: $\theta = \tan^{-1}(120/50) = \tan^{-1}(2.4) \approx 67.4^\circ$. Connect to the driving question: either direction of the calculation is useful — knowing the

	do you find the angle?' Pedagogical note: Insist students set calculators to DEGREE mode first; wrong mode is the single most common calculator error in trigonometry.		angle lets you find the height; knowing both sides lets you find the angle.
Lesson 6 Complementary Angles and the $\sin \theta = \cos(90^\circ - \theta)$ Identity	Teacher asks students to evaluate $\sin 30^\circ$ and $\cos 60^\circ$ on their calculators, then $\sin 25^\circ$ and $\cos 65^\circ$, then $\sin 72^\circ$ and $\cos 18^\circ$. Students record the pairs and observe that each pair gives identical values. Pedagogical note: After students notice the pattern, give 60 seconds of silent writing time — 'Write in your own words why you think this happens before we discuss it as a class.' Cold-call three students.	Students discovered and proved that $\sin \theta = \cos(90^\circ - \theta)$ for any acute angle θ , because in a right-angled triangle the two acute angles are complementary (sum to 90°), so the side that is Opposite for θ is the Adjacent for the other angle. The identity is not a coincidence but a geometric inevitability.	Label the two acute angles of a right-angled triangle as θ and $(90^\circ - \theta)$. The side Opposite θ is Adjacent to $(90^\circ - \theta)$, and both share the same Hypotenuse. Therefore $\sin \theta = \text{Opp/Hyp} = \cos(90^\circ - \theta)$. Teacher should note the word origin: 'cosine' literally means 'complement's sine'. Practical implication for Kenya Certificate of Secondary Education (KCSE) exams: this identity allows simplification of expressions and is often tested. Example: $\sin 67^\circ = \cos 23^\circ$ — useful when a table only lists angles up to 45° .
Lesson 7 Special Angles (Derivation): Unlocking Exact Trigonometric Values	Teacher draws an equilateral triangle with side 2 and bisects it to produce a 30-60-90 triangle, then draws an isosceles right-angled triangle with legs of length 1. Students measure and compute ratios, then compare with calculator values for 30° , 45° , 60° . They observe that the geometric derivation gives exact values (e.g., $\sin 60^\circ = \sqrt{3}/2$) while calculators give decimal approximations. Pedagogical note: Ask 'Which answer is more precise — the calculator's 0.866 or the exact $\sqrt{3}/2$?' Allow debate before explaining that surds are exact and decimals are rounded.	Learners learned that five special angles (0° , 30° , 45° , 60° , 90°) have exact trigonometric ratios derivable from geometry alone — no calculator needed. These values (involving 1, $\sqrt{2}$, $\sqrt{3}$, and 2) recur throughout mathematics and physics and must be memorised for KCSE. The derivation from an equilateral triangle (side 2) and a unit isosceles right triangle gives students a method to reconstruct the table if memory fails.	Teacher should display the complete special-angles table and walk through each derivation step. For 30-60-90: half the equilateral triangle gives sides 1, $\sqrt{3}$, 2, so $\sin 30^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$, $\tan 30^\circ = 1/\sqrt{3} = \sqrt{3}/3$. For 45-45-90: unit legs give hypotenuse $\sqrt{2}$, so $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2} = \sqrt{2}/2$, $\tan 45^\circ = 1$. Boundary values: $\sin 0^\circ = 0$, $\cos 0^\circ = 1$; $\sin 90^\circ = 1$, $\cos 90^\circ = 0$. Connect to driving question: a sunbeam at exactly 30° elevation means height = $\tan 30^\circ \times \text{shadow} = (\sqrt{3}/3) \times \text{shadow}$ — an exact, elegant answer.
Lesson 8 Special Angles: Solving Real Problems Without a Calculator	Teacher presents three problems set in Kenyan contexts: (1) A mobile phone mast in Nakuru casts a 20 m shadow when the sun's elevation is 60° — find the mast's height. (2) A ramp at a Mombasa port makes a 45° angle with the ground and has a horizontal run of 5 m — find its vertical rise. (3) A Maasai elder in the Rift Valley observes the sun at 30° elevation and his walking stick's shadow is 1.2 m — find the stick's length. Students solve without calculators using exact values. Pedagogical note: Circulate and watch for students converting exact surds to decimals	For special angles, exact trigonometric ratios allow clean, calculator-free solutions. Students practised the full problem-solving sequence: (1) draw and label the right-angled triangle, (2) identify θ and the known/unknown sides, (3) select the correct ratio (SOH–CAH–TOA), (4) substitute exact values, (5) solve algebraically, (6) interpret the answer in context.	Exact answers are preferred in mathematics because they carry no rounding error and reveal structure. Teacher should show that the mast problem gives height = $\tan 60^\circ \times 20 = \sqrt{3} \times 20 = 20\sqrt{3} \text{ m} \approx 34.6 \text{ m}$, and contrast this with the rounded decimal to reinforce why exact form is valued. Pedagogical note: Reinforce that the six-step sequence used here is identical to the one used with calculator-dependent angles; special angles simply replace the calculator lookup with a memorised exact value.

	prematurely; redirect them to leave answers in surd form until the final step.		
Lesson 9 Reciprocal Trigonometric Ratios: Cosecant, Secant, and Cotangent	Students compute $\operatorname{cosec} 30^\circ$, $\sec 45^\circ$, and $\cot 60^\circ$ from their tables and verify using a calculator ($1 \div \sin$, $1 \div \cos$, $1 \div \tan$). They observe that $\operatorname{cosec} \theta \geq 1$ and $\sec \theta \geq 1$ for all acute angles, while $\cot \theta$ decreases as θ increases (mirroring $\tan \theta$'s increase).	The three reciprocal ratios are: $\operatorname{cosec} \theta = 1/\sin \theta = \text{Hypotenuse/Opposite}$; $\sec \theta = 1/\cos \theta = \text{Hypotenuse/Adjacent}$; $\cot \theta = 1/\tan \theta = \text{Adjacent/Opposite}$. These extend the trigonometric toolkit and appear in KCSE problems, physics (wave equations), and higher mathematics. Students learned to evaluate them for special angles and to use the identity $\cot \theta = \cos \theta / \sin \theta$.	Reciprocal ratios arise naturally when a problem gives the Hypotenuse and asks for the Opposite (use cosec) or gives the Adjacent and asks for the Hypotenuse (use sec). Teacher should present a worked example: a lighthouse keeper in Mombasa knows the beam travels 200 m (Hypotenuse) at 40° elevation — the height above sea level = $\sin 40^\circ \times 200$, but equivalently height = $200 / \operatorname{cosec} 40^\circ$. Both formulations are equivalent; the reciprocal form is useful when equations are rearranged algebraically. Pedagogical note: Students often confuse $\cos^{-1}$ (inverse cosine, an angle) with sec (reciprocal cosine, a ratio); draw the distinction explicitly and repeatedly.
Lesson 10 Real-Life Applications, DQB Closure & Final Model Revision	Teacher returns to the original driving question photograph from Lesson 1 (acacia tree, Rift Valley). Students are asked to fully solve the problem using everything learned across the strand: identify the right-angled triangle, label sides, choose the appropriate ratio (tan for height/shadow, sin or cos if hypotenuse is involved), compute the answer, and state it in context. The Driving Question Board is reviewed — every question posted across 10 lessons is addressed and answered. Pedagogical note: Give students 5 minutes of silent individual work before pair-share, then cold-call three students to present solutions to the class.	Students learned that the trigonometric ratios (sin, cos, tan and their reciprocals cosec, sec, cot) form a unified, interconnected system — not isolated formulas. The sub-strand equips learners with a precise mathematical language for describing angles and indirect measurement in any right-angled triangle. Applications span tree and building heights, hill gradients, navigation, architecture, and engineering across Kenya.	The driving question is fully answered: the angle a sunbeam makes with the ground (angle of elevation) is θ in a right-angled triangle whose Opposite side is the height of the object and whose Adjacent side is the shadow length. Therefore height = $\tan \theta \times \text{shadow length}$ — and this works for any tree in Karura Forest, any building along Nairobi's skyline, or any hill in the Aberdare Range, without ever measuring the height directly. Teacher should close by having students write a one-sentence answer to the driving question in their own words as an exit ticket, confirming that the cumulative model built over 10 lessons is complete and coherent.